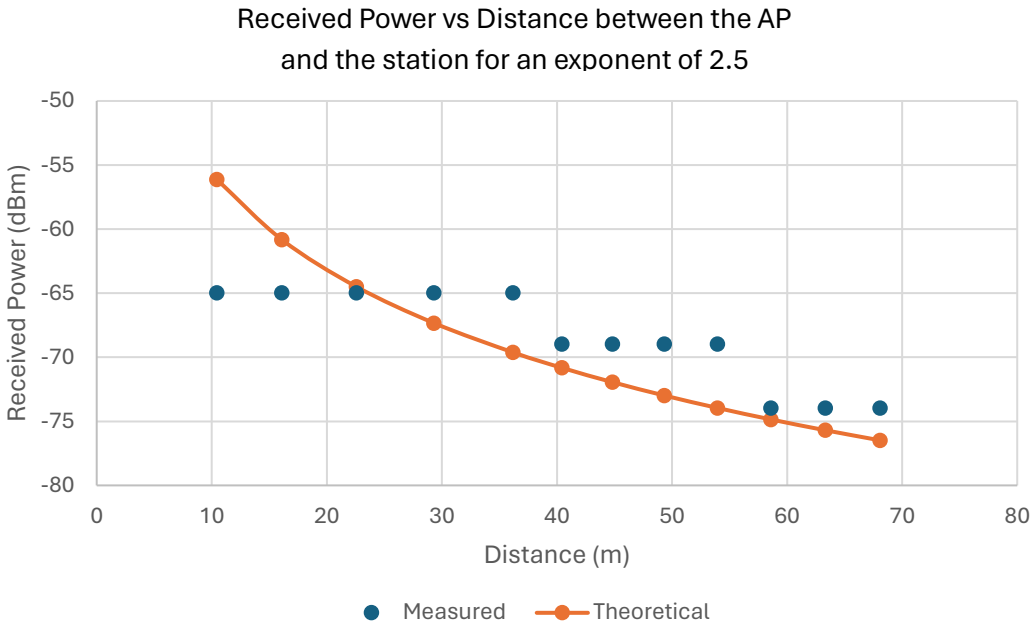
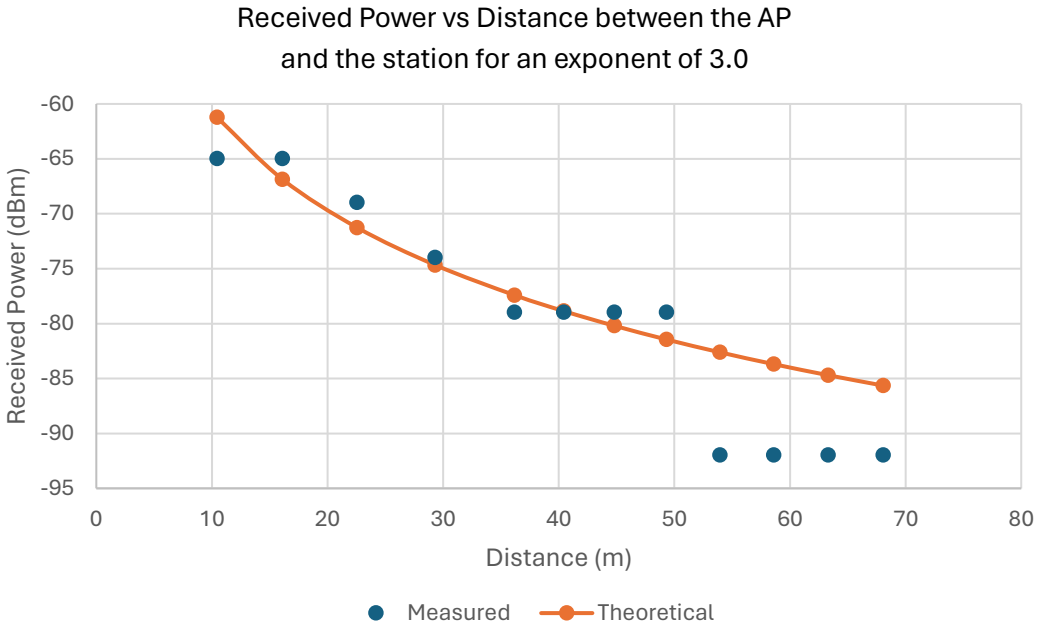
Félix Arnold

ECE 6110 Lab Assignment 4
Wireless Propagation and Rate Adaptation

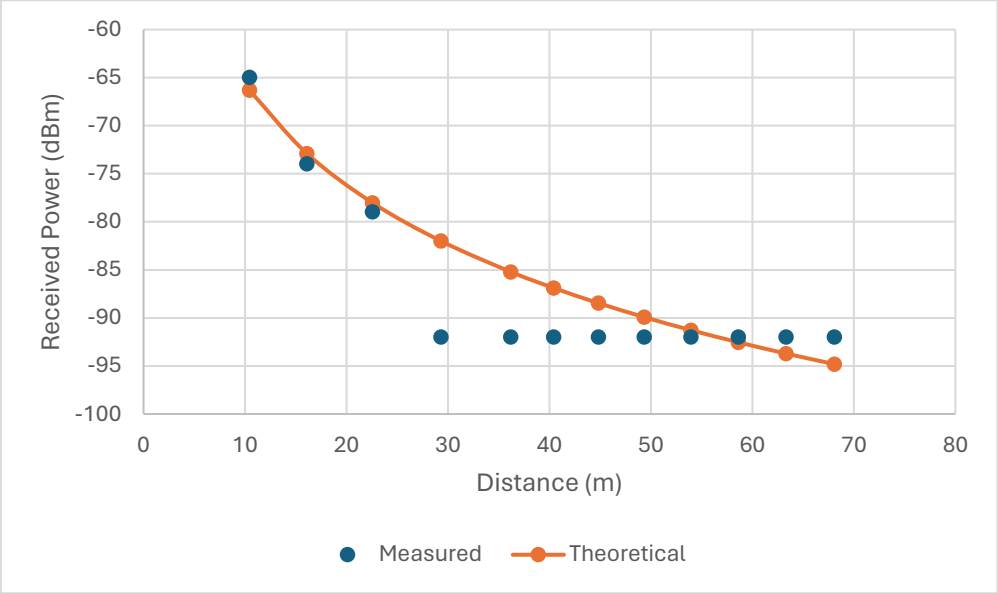
Part 1: Evaluation of Different Path Loss Exponents

In this project, I set up a WiFi network simulation in order to evaluate the impact of the path loss exponent on signal-to-noise ratio by varying the distance between an access point and a wireless station. The simulation uses the 802.11ac standard, a 20 MHz channel width, and the MinstrelHtWifiManager for rate control. The system is composed of a access point and a single wireless station, whose position is updated every 2 seconds.





Received Power vs Distance between the AP
and the station for an exponent of 3.5



We observe for all 3 plots that the measured and theoretical curves somewhat follow the same trajectory. As expected, as the path exponent increase, we observe a quicker decrease of the received power. The main difference is that while the theoretical curve of the received power decreases smoothly, the measured one using Minstrel shows stages during which the received power remains constant before dropping to another stage.

This is due to the Minstrel algorithm. Its role is to adapt the rate of transmission between a sender and a receiver, in our case this is necessary as the AP and the station are getting further and further away, meaning that the less and less data are able to being received. Without the Minstrel algorithm, a high data rate would cause an important packet drop rate, thus causing a worse throughput compared to a lower data rate.

The minstrel algorithm works by computing an empirical probability of successful transmission on a fixed amount of available rate. It will then use these probabilities to select the most appropriate rate. As the values of the chosen data rates are fixed, the received power will remains constant while this rate is deemed the most appropriate one and, in our case, will only change when the distance is big enough for another data rate to be more suitable.